Response to the reviewer

A reliability-constrained, multi-objective framework for minimum-mass design of steel frame structures under material and loading uncertainty

[Author name]

We thank the reviewer for a careful and constructive report. We have revised the software as well as the manuscript. Every number in the revised manuscript is regenerated from the seeded scripts in the public repository, and the earlier results were rerun, not edited. The reviewer's comments are summarised in italics, followed by our response and the changes made. Section, table, figure and equation numbers refer to the revised manuscript.

Summary of the main changes

1. A *code-compliant* formulation, RBDO-C, now imposes the reliability targets *and* all deterministic AISC 360-16 strength, drift and deflection checks (Section 2.6). It is the main reliability-based formulation throughout the paper. The reliability-only formulation (RBDO) is kept for comparison.
2. Every reliability estimate is reported with its failure count, an exact (Clopper–Pearson) 95% confidence interval for P_f and β, and a three-way acceptance rule (Section 2.5, Eq. (6)).
3. A literature comparison table with eight closely related studies and a precise statement of what is new (Section 1, Table 1).
4. The surrogate is validated against full second-order re-analysis for eight designs. The validation includes 600 random samples and up to 500 targeted samples near the limit state per design, and it gives a confidence interval for the difference in β (Section 3.3, Table 7).
5. A separate treatment of model (epistemic) uncertainty with frame-wide model factors (Section 2.4, Table 19).
6. Optimiser robustness: 2–10 seeds per formulation and frame, SAA sizes from 2×10^4 to 10^5 with three independent streams, and an exact enumeration of a reduced problem with 11 200 designs (Section 4.1).
7. Deterministic optima with drift limits $h/400$–$h/600$, section counts and an illustrative cost index (Sections 4.4 and 4.7).
8. A new title ("minimum-mass"), a rewritten abstract, a reproducibility statement with a one-command reproduction script, and consistent terminology for the ULS.

Major comments

Comment 1 (code compliance). *The RBDO optimum of the 6-m bay frame has an LRFD strength ratio of 1.011. State whether the formulation includes the AISC checks, provide a code-compliant RBDO and report whether the conclusions change.*

Response. The original RBDO formulation imposed only the reliability targets (ULS, drift and deflection) and the column-continuity rule. It did not impose any deterministic check, including the LRFD strength check. The revised manuscript says so explicitly (Section 2.6). We added RBDO-C, which additionally requires every deterministic check (LRFD strength ratio ≤ 1, drift $\leq h/400$, deflection $\leq L/360$). We reran it for all seven frames with at least two GA seeds each and verified every optimum with 10^6 independent samples (Table 14).

The code checks are active in two of the seven frames. The 6-m bay RBDO design (LRFD ratio 1.011) is replaced by an RBDO-C design with ratio 0.914, which is 0.32% heavier. The low-wind RBDO design (ratio 1.002) is replaced by one that is 0.64% heavier. In the other five frames RBDO and RBDO-C give the same design. All RBDO-C designs satisfy every LRFD check and meet both targets under the acceptance rule. The qualitative conclusions do not change. The mass premium of RBDO-C over DDO is +1.13% to +10.11% (+2.88% for the base frame). It is driven by the 50-year drift reliability of the deterministic optima ($\beta_{drift} = 0.89$–2.54). Section 5 explains how RBDO-C fits the code: it is a least-mass code-compliant design that also has a stated reliability.

Changes. Sections 2.6, 4.2, 4.5; Tables 11, 14; Fig. 11; all figures and tables now use RBDO-C.

Comment 2 (statistical treatment). *Distinguish point estimates from confidence bounds, give binomial confidence intervals for P_f and β, explain acceptance of designs near the threshold, and report failure counts. The "0.5/N lower bound" wording is not justified.*

Response. We agree. Section 2.5 now defines the point estimate $\hat{P}_f = k/N$ and the exact Clopper–Pearson interval (Eq. (6)), with its mapping to β. For zero failures it uses the one-sided bound: with $N = 10^6$, $P_f \leq 3.0 \times 10^{-6}$ and $\beta \geq 4.53$. The "0.5/N" wording has been removed.

Each verified design is reported with N, k, $\hat{\beta}$ and the 95% interval (Tables 11, 12, 13, 14). A design is classified as *above*, *within* or *below* the target, depending on whether the interval lies above the target, contains it or lies below it. "Within" is accepted, because the target cannot be resolved more finely at this sample size. This is stated as a convention, and its consequences are discussed in Section 5. Only one final design is "within": the 3-storey RBDO-C design, with $\beta_{ULS} = 2.995$ [2.98, 3.01]. The others are above both targets.

We also report the stricter alternative. In the base frame, imposing the target on the one-sided lower confidence bound inside the SAA costs 1.6% more steel (Table 10). Section 2.5 explains the risks of common random numbers (optimisation bias and correlated estimates) and how we control them: independent re-verification, other SAA sizes and streams, and the lower-bound variant.

Changes. Section 2.5; Tables 11, 10, 12, 13, 14; Figs. 9, 11, 10.

Comment 3 (novelty). *Add a comparison table of 5–10 close studies and state precisely what is new.*

Response. Table 1 compares eight studies on reliability-based or probabilistic optimisation of steel frames, ordered by year, with our study. Its columns are the structural model, design variables, reliability formulation, optimiser, verification and reproducibility. The introduction then states the contribution in four points. We do not claim novelty for any single ingredient. The new elements are:

(i) a like-for-like comparison of DDO, DDO with tightened drift limits, RBDO and code-compliant RBDO on the same frames, catalogue and optimiser, with interval-verified reliability;

(ii) serviceability reliability treated together with a system ULS;

(iii) a verification chain that ends in a stratified full second-order Monte Carlo validation for several designs;

(iv) an open release with a one-command reproduction.

The table entries are based on the abstracts and bibliographic records available to us; "n.r." means not reported there. We have checked the bibliographic data of the six references added to the reference list.

Changes. Section 1, Table 1; six references added.

Comment 4 (validation breadth). *Validate several designs (9 storeys, 10-m bays, high wind), compare failure classifications, report the uncertainty of the difference in P_f and show the accuracy near the constraints.*

Response. Section 3.3 now validates eight designs: the base DDO design and the RBDO-C designs of all seven frames, including the 9-storey, 10-m bay and high-wind frames. For each design it uses two sets of samples:

(a) 600 random samples of the wind scenario, which give the bias and scatter of the utilisation and drift and the agreement of the drift classification;

(b) a stratified tail estimator. From a pool of 2×10^5 samples, all samples with surrogate utilisation $U \geq 1$ (up to 250) and 250 samples with $0.85 \leq U < 1$ are re-analysed with a full P-Δ PyNite model with member-wise random modulus and fabrication factors. This gives P_f of the full model, its ratio to the surrogate and the difference in β, with Bayesian (Jeffreys) 95% intervals.

The results are in Table 7 and Fig. 3:

- The utilisation bias is +0.4 to +1.1% (s.d. 1.0–1.6%) and the drift bias is −1.0 to +1.4%.
- 0–7 of the 600 drift classifications disagree.
- 79–91% of the surrogate failures also fail in the full model, and 0–4 of the 250 near misses fail in the full model.
- $\Delta\beta = \beta_{full} - \beta_{sur}$ lies between −0.07 and +0.05, and its interval contains zero for seven of the eight designs.

The exception is the 3-storey RBDO-C design. There, $\Delta\beta = -0.07$ [−0.17, −0.01]; the surrogate is slightly unconservative and the design's full-model β_{ULS} is about 2.93. We report this and discuss it rather than hiding it. Accuracy near the constraints is shown by the near-miss stratum and by the largest full-to-surrogate utilisation ratio (at most 1.049).

Changes. Section 3.3; Table 7; Figs. 3, 4; new function `broad_validation` and stage `validate`.

Comment 5 (types of uncertainty). *Separate aleatory from epistemic and model-form uncertainty, justify the dependence assumptions, discuss the omissions (residual stresses, imperfections, connections, inelastic redistribution) and limit the interpretation of the system ULS.*

Response.

- *Uncertainty types.* Section 2.4 now distinguishes the aleatory variables of Table 3 from model-form uncertainty. Model-form uncertainty is represented by frame-wide lognormal factors on resistance, load effect and drift. Because the baseline indices are conditional on a perfect model, these factors are *not* part of the baseline. Their effect is quantified separately (Table 19): a c.o.v. of 0.05 lowers β_{ULS} by 0.23–0.30, a c.o.v. of 0.10 lowers it by 0.82–0.90, and β_{drift} changes by at most 0.05.

- *Dependence.* Section 2.3 explains how the system event is built: the union over all members and both Turkstra scenarios. Dependence enters through the shared loads, modulus, partly correlated yield stresses ($\rho = 0.5$, assumed), fabrication factor and imperfection, and it is represented exactly by joint sampling. Section 4.8 tests the assumed values.
- *Omissions and interpretation.* We list the omissions (residual stresses and member imperfections only through the AISC column curves and τ_b; no inelastic redistribution; rigid, non-failing connections). The ULS is defined consistently as "first attainment of a member limit state in a second-order elastic analysis, not collapse" in the abstract, Section 2.3 and the Discussion.

Changes. Sections 2.3, 2.4, 4.8, 5; Table 3 (new Basis/Status columns and a model-factor row); Table 19; Fig. 12.

Comment 6 (robustness of the optimiser). *Use more seeds; report best, median and worst results; count distinct designs; use an alternative search or reduced-space enumeration; state the stopping criterion; show the effect of the SAA size on feasibility; do not claim global optimality.*

Response.

- *Seeds.* Table 8 reports best, median and worst masses, the number of distinct final designs and the number of designs evaluated for every frame and formulation. The base frame was run with 10 seeds (DDO) and 5 seeds (RBDO-C); the other frames with 2–3 seeds. For the base frame, 8 of 10 DDO runs found the best design (the other two were 0.3% heavier) and all five RBDO-C runs found the same design. The largest spread is 1.6%, for the high-wind RBDO-C runs, and we report the lightest.
- *Enumeration.* A reduced problem with three variables and 11 200 designs was solved exactly by enumeration in order of mass (Table 9). The GA found the exact optimum in 10 of 10 runs for each formulation, after evaluating about 700 designs.
- *SAA size.* $N = 2 \times 10^4$, 5×10^4 (three independent streams) and 10^5 all give the same base-frame RBDO-C design (Table 10).
- *Stopping and optimality.* The stopping criterion is stated in Section 2.6. We no longer claim global optimality; the reported designs are the best found.

Changes. Sections 2.6, 4.1; Tables 8, 9, 10; Fig. 6.

Comment 7 (economics). *Use mass as the objective and cost only as a proxy; report section counts; add a simple cost sensitivity; examine several drift limits, not only $h/500$.*

Response. The objective is now called steel mass throughout, including the title. Cost is discussed only through an illustrative cost index (Section 2.7, Eq. (12)). The index adds a per-section penalty and a beam-mass term for connection work, evaluated for five parameter pairs. Table 18 gives the number of distinct sections and the index. The ranking of DDO and RBDO-C changes in only one case (6-m bays with a high section penalty, -0.19%).

Deterministic optima were computed for drift limits $h/400$, $h/450$, $h/500$ and $h/600$ for every frame (Table 13, Fig. 10). The lightest deterministic design that meets both targets is $h/500$ in four frames and $h/600$ in two. In the 3-storey frame none of these limits is enough, because that frame is strength-governed. RBDO-C is within -5.0% to $+0.6\%$ of that design. It is $+0.58\%$ heavier in the 6-m bay frame, because the $h/500$ design meets the drift target only within the confidence interval and was rejected by the SAA estimate. The manuscript reports this.

Changes. Title; Sections 2.7, 4.4, 4.7; Tables 13, 18; Fig. 10.

Additional technical issues

Comment T1 (B_1–B_2). *Explain how B_1 and B_2 are recomputed and what is assumed.*

Response. Section 2.2 now states that B_1 and B_2 are recomputed for every sample: P_{story}, P_r and C_m follow the sampled gravity loads, and $P_{e,story}$ and P_{e1} follow the sampled modulus factor. It lists four assumptions:

(a) storey stiffness scales with the frame-wide modulus factor;

(b) gravity load is distributed as in the first-order analysis;

(c) there is no stiffness loss before the first member limit state;

(d) the storey amplification is the same for all columns of a storey.

Assumptions (a) and (b) are removed in the full re-analysis of Section 3.3, and (c) is part of the elastic ULS definition. The comparison with OpenSeesPy and PyNite P-Δ (Table 5, Fig. 2) shows that the amplified drifts are within 1.8% of the second-order solution. We corrected the earlier statement that they were "always conservative": the surrogate underestimates the drift by up to 1.0%.

Comment T2 (dependence and system event). *Explain the dependence between scenarios and members and how the system event is built.*

Response. See the response to Comment 5 and Section 2.3. We also report the share of failures by member type and scenario (Table 15).

Comment T3 (distribution table). *Separate sourced parameters from assumed ones.*

Response. Table 3 now has a "Basis" column with the source and a "Status" column (sourced / assumed / adapted). The assumed values (bias of the yield stress 1.10, correlation 0.5, fabrication c.o.v., model factors) are flagged and tested in Section 4.8.

Comment T4 (common random numbers). *Discuss the risks of using common random numbers.*

Response. Section 2.5; see the response to Comment 2.

Comment T5 (unrounded values, counts, intervals, tolerance).

Response. Percentages are now computed from unrounded masses (in kg). Indices are given to three decimals with failure counts and intervals. The acceptance tolerance is the confidence interval itself (Section 2.5).

Comment T6 (load paths and member-level results for more frames).

Response. Table 15 reports, for all seven frames and both DDO and RBDO-C, where the failures occur (beams, exterior or interior columns) and in which scenario. Section 4.5 explains where RBDO-C adds steel, e.g. +577 kg in beams and −151 kg in columns for 10-m bays. Fig. 8 shows member utilisations of the base frame.

Comment T7 (reproducibility). *Give the repository URL, commit, environment and a one-command reproduction.*

Response. The "Data and code availability" statement gives the URL, folder, branch and environment (`requirements.txt`). The script `reproduce.sh` reruns every stage and rebuilds the manuscript. The exact commit is recorded in the repository. We will create a release tag with a DOI on acceptance.

Terminology, figures and numerical consistency

Comment ULS terminology.

Response. "Component-based, second-order elastic ULS (first attainment of a member limit state, not collapse)" is now used consistently in the abstract, Section 2.3, the Discussion and the Conclusions.

Comment Fig. 2 (P-Δ comparison). *State where the largest differences are.*

Response. The caption of Fig. 2 and the text in Section 3.2 now say that the largest differences occur in the upper storeys and that the surrogate underestimates the drift by at most 1%.

Comment Fig. 3 (full Monte Carlo). *Add classification statistics and confidence intervals.*

Response. These statistics are now in Table 7 and Fig. 3. The caption of Fig. 4 gives the sample sizes.

Comment Fig. 8 (Pareto front). *Separate verified designs from SAA estimates and show confidence intervals.*

Response. In Fig. 9 the non-dominated designs are re-verified with 4×10^5 independent samples and shown with 95% intervals. Designs that fail the LRFD check are drawn with open markers. SAA-only values are drawn in grey, and the DDO and RBDO-C optima (10^6 samples) are marked separately.

Comment Numerical audit. *The 2.99 index, the 1.011 ratio, counts consistent with β, percentages from unrounded masses, and consistent notation.*

Response.

- The index 2.99 is the verified β_{ULS} of the reliability-only (RBDO) low-wind design. It is now reported unrounded with its count and interval: 2.986 [2.97, 3.00] ($k = 1415$ in $N = 10^6$). This interval contains the target, so the design is "within" under the acceptance rule. The code-compliant design for this frame has 3.019 [3.00, 3.04] (Table 14).
- The 1.011 ratio is explained under Comment 1.
- The tables are generated by `make_figures.py` directly from the result files, so counts, indices and intervals are consistent by construction.
- Percentages are computed from unrounded masses.
- Notation: P-Δ denotes the storey (sway) effect and P-δ the member effect; both are used consistently.

Minor comments

- *Title.* Changed to "... minimum-mass design...".
- *Abstract.* Rewritten. It states the component-based second-order elastic ULS and the model-uncertainty treatment, and it contains fewer numbers.
- *Author placeholders.* The author names and affiliations remain placeholders in this version and will be completed before resubmission.
- *Reproducibility statement.* See T7.
- *Units and notation.* See the numerical audit above.

- *References.* Six references were added, with bibliographic data checked against publisher or indexing records.
- *Figures and captions.* All figures were regenerated from the result files. Legends were moved so that they do not overlap the data, and the captions state the sample sizes.
- *Repetition.* The Results and Discussion were restructured to state each finding once.

Limitations acknowledged in the revision

- The surrogate is slightly unconservative for the 3-storey design ($\Delta\beta = -0.07$).
- RBDO-C is 0.58% heavier than the $h/500$ deterministic design in the 6-m bay frame.
- The GA spread is 1.6% for the high-wind RBDO-C runs.
- The literature table relies on abstracts and records.
- The conclusions are limited to regular planar frames under gravity and wind.

These limitations are stated in the manuscript.